

CLASS TOPIC

Hypothesis testing II - two samples and chi-square

What this class fixes

- Know the central question of the topic.
- Use the correct notation and units.
- Avoid mechanical calculation without interpretation.

W11**Statistics**

Clean formulas · aligned layout · corrected te

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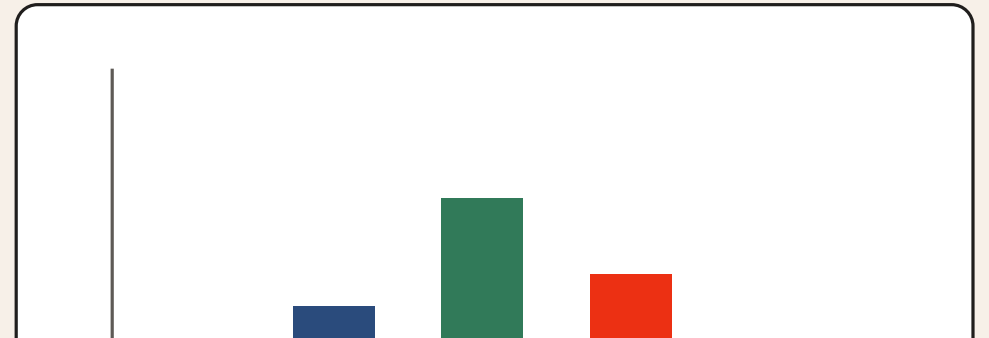
Hypothesis testing II - two samples and chi-square

Economic question first

- Start from a real decision: compare, describe, forecast or test.
- Translate the question into variables before using formulas.
- Define the population, sample and measurement unit.

FORMULA

$$\text{two means: } (\bar{x}_1 - \bar{x}_2) \pm t^* \cdot \text{SE}$$
$$\text{SE} = \sqrt{s_1^2/n_1 + s_2^2/n_2}$$



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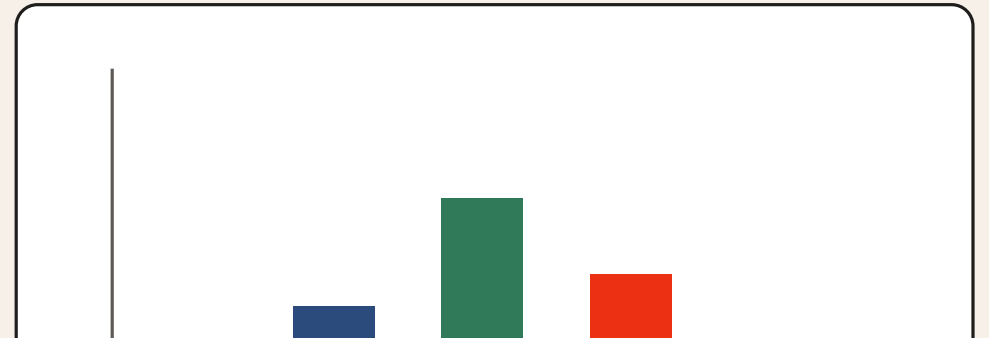
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Key definitions

- Population: full group of interest.
- Sample: observed part of the population.
- Statistic: number computed from a sample;
parameter: number for the population.

FORMULA

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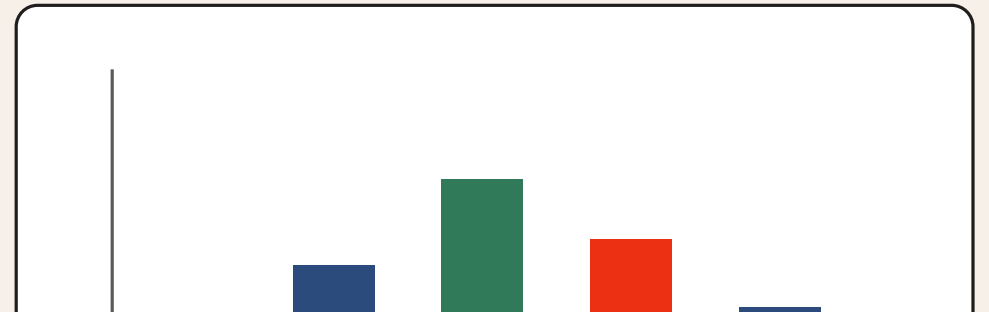
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Main formula box

- All formulas are kept in one merged formula area on this slide.
- Read every symbol before calculating.
- Use parentheses carefully.

FORMULA 1

$$\text{two means: } (\bar{x}_1 - \bar{x}_2) \pm t^* \cdot SE$$
$$SE = \sqrt{s_1^2/n_1 + s_2^2/n_2}$$



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Mini example

- Write the data clearly.
- Substitute values step by step.
- Round only at the final reporting stage.

01 Data: 4, 7, 8, 9, 12

02 Step 1: compute the needed statistic

03 Step 2: write the result with units.

04 Step 3: explain it in one sentence.

FORMULA

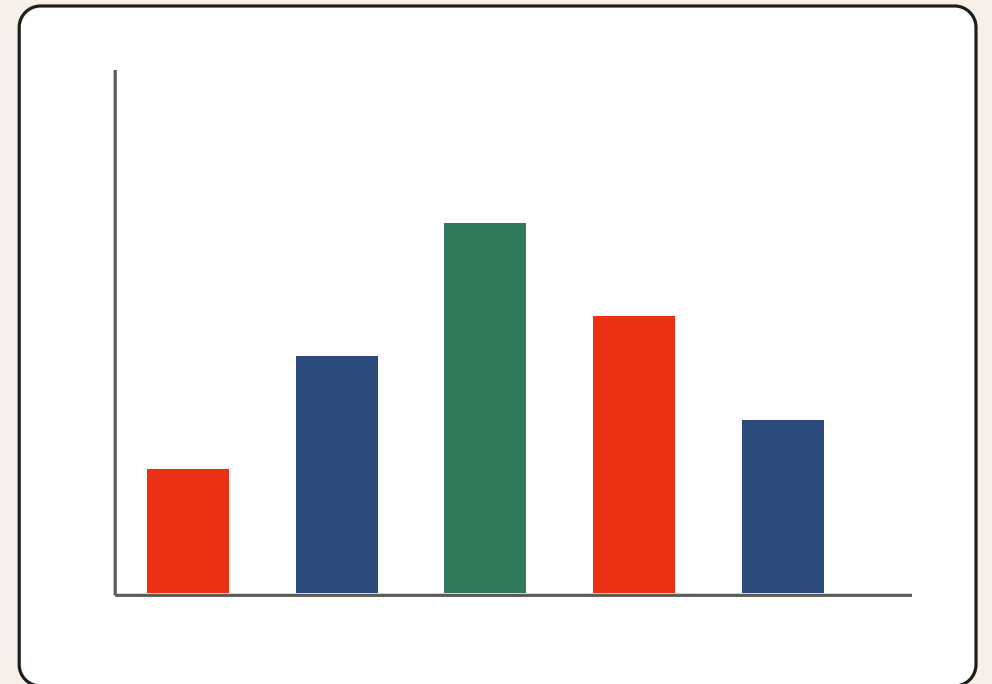
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Visual intuition

- A graph should answer a question, not decorate the slide.
- Always label axes and units.
- Look for center, spread, shape and unusual values.



FORMULA

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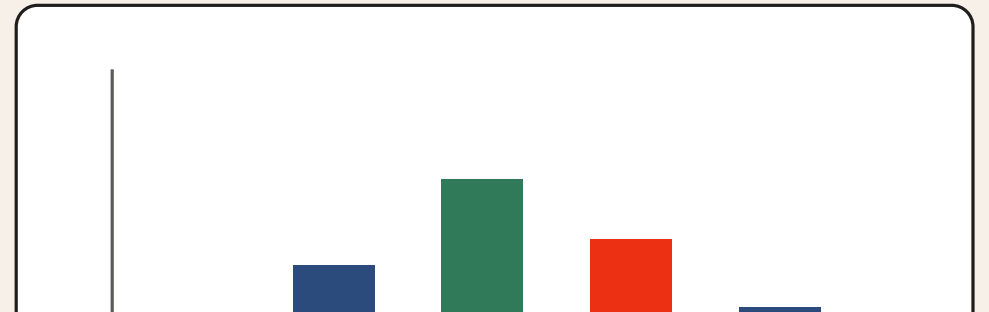
Second formula box

- This formula is separated from explanations but kept in one clean box.
- No broken equation fragments or floating text boxes.
- Notation is consistent across English and Russian versions.

FORMULA 2

$$\chi^2 = \sum (\text{Observed} - \text{Expected})^2 / \text{Expected}$$
$$\text{df} = (\text{rows} - 1)(\text{columns} - 1)$$

Formula boxes fixed: one merged box, consistent symbols, no broken fragments.



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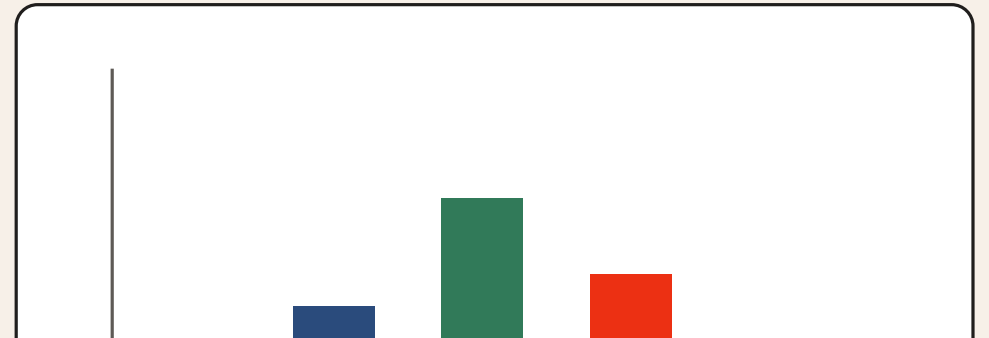
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Common mistakes

- Mixing sample and population symbols.
- Using a formula without checking conditions.
- Reporting a number without units or interpretation.

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R / spreadsheet check

- Use software to calculate, but keep the statistical logic visible.
- Check variable names, missing values and units first.
- Reproducible work is easier to grade and correct.

R / EXCEL

```
# example pattern  
mean(x); sd(x); summary(x)  
plot(x, y); cor(x, y)
```

FORMULA

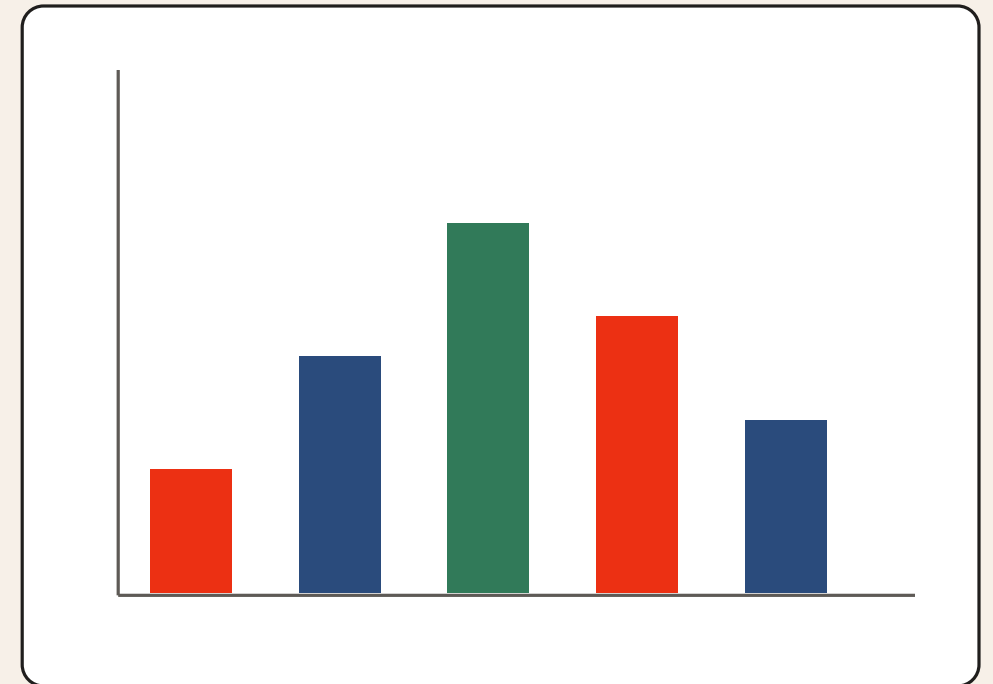
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Interpretation sentence

- Do not stop at the number.
- Explain what the result means in the original context.
- Use cautious language when evidence is limited.



FORMULA

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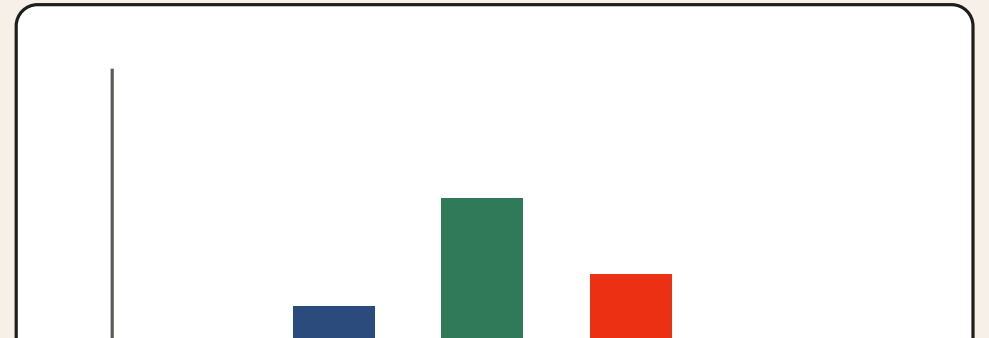
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Conditions and assumptions

- Check whether the method fits the data type.
- Check independence and sample size when needed.
- If assumptions fail, explain the limitation.

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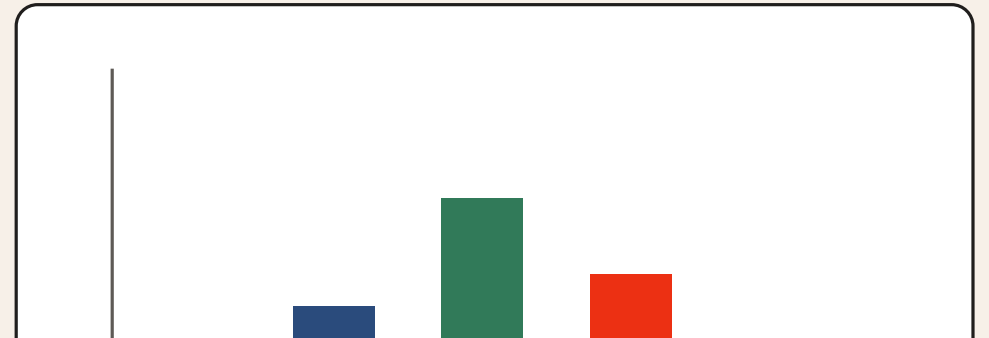
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Class exercise

- Solve one calculation by hand.
- Then verify it with software.
- Write a two-sentence conclusion.

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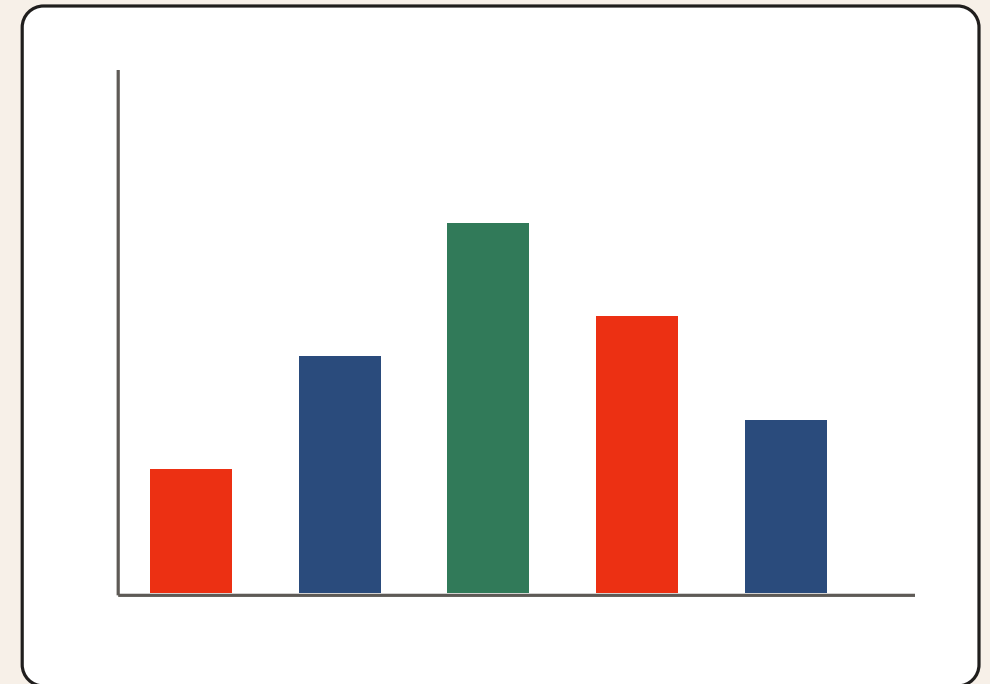


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How to report

- Report method, estimate, uncertainty and interpretation.
- Use tables only when they make comparison easier.
- Separate statistical evidence from causal claims.

**FORMULA**

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Final checklist

- Question is clear.
- Variables and units are defined.
- Formula is correct.
- Result is interpreted.
- Limitations are stated.

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